METRIC

MIL-C-24621B
6 June 1991
SUPERSEDING
MIL-C-0024621A(NAVY)
16 May 1989
DOD-C-24621
31 August 1984

MILITARY SPECIFICATION

COUPLERS, PASSIVE, FIBER OPTIC, GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and characteristics of passive fiber optic couplers (see 6.4.7) intended to optically interconnect three or more optical fibers. Directional and bidirectional couplers are covered by this document. Couplers are suitable for interconnection to cable fibers as specified in DOD-C-85045. They are also compatible with splices as specified in MIL-S-24623, and connectors as specified in MIL-C-28876 and MIL-C-83522 (see 6.4.7).

1.2 Classification. Couplers are classified as specified in 1.2.1 through 1.2.3.

1.2.1 Type. The type designation defines the wavelength of operation of the coupler as well as the method of interfacing the coupler to external fiber optic cables for system applications.

a. Type 1: Nominal center of operation at 1.31 micrometers (µm), with a range of 1.26 to 1.36 µm. Fiber optic cable pigtails are an integral part of the coupler assembly.

b. Type 2: Nominal center of operation at 1.55 µm, with a range of 1.50 µm to 1.60 µm. Fiber optic cable connectors are an integral part of the coupler assembly housing.

c. Type 3: Wavelength independent (range of operation that includes part of all of the wavelength ranges if types 1 and 2).

1.2.2 Class. The class designation defines the optical power distribution made by the coupler. The tolerances of the elements of the transmittance matrix that allow the coupler to be classified as uniformly distributive or nonuniformly distributive (see 6.4.6 and 6.4.10) will be indicated in the specification sheet (see 3.1).

a. Class A: Uniformly distributive (see 6.4.10).

b. Class B: Nonuniformly distributive (see 6.4.6).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command (SEA 55Z3), DOD Standardization Program and Documents Division, Department of the Navy, Washington DC 20362-5101, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Style. The style designation defines the style of packaging.

a. Style 1: Couplers with single mode fiber pigtails.

b. Style 2: Couplers with multi-mode fiber pigtails.

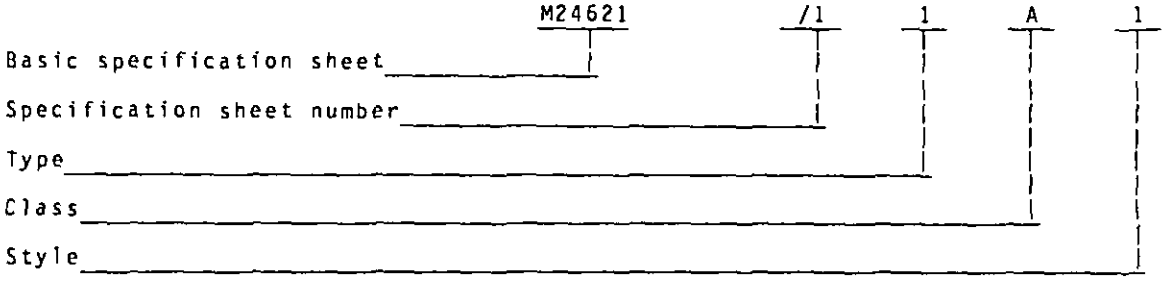
c. Style 3: Couplers with single mode cable pigtails.

d. Style 4: Couplers with multi-mode cable pigtails.

e. Style 5: Couplers with single mode fiber optic connectors as part of coupler housing.

f. Style 6: Couplers with multimode fiber optic connectors as part of coupler housing.

1.3 Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term (part number, identification number, and type designator) which was previously used in this specification. The couplers specified herein (see 3.1) shall be identified by PINs which shall consist of the basic specification number, slash sheet of the specification sheet, and a sequentially assigned number as shown in the following example:



2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-I-735	-	Isopropyl Alcohol.
QQ-S-766	-	Steel Plate, Sheet, and Strip-Corrosion Resisting.

MILITARY

MIL-S-901	-	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-G-3056	-	Gasoline, Automotive, Combat.
MIL-S-5002	-	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems.
MIL-H-5606	-	Hydraulic FLuid, Petroleum Base, Aircraft, Missile, and Ordinance.

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MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-I-17214	-	Indicator, Permeability, Low-mu (go-no-go).
MIL-L-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.
DOD-C-24621/1(SH)	-	Coupler, Fiber Optic Cable, Cable Splitter, Passive Glass-Connectorized Output; Type I 820 to 910 Nanometers Wavelength Range, 2 X 2 Symmetrical Transmission Star, 50/125 Micrometer Fiber, Graded Index Multi-Mode Bidirectional (Metric).
DOD-C-24621/2	-	Coupler, Fiber Optic Cable, Cable Splitter, Passive, Glass-Pigtailed Output; Type II 820 to 910 Nanometers Wavelength Range, 4 X 4 Symmetrical Transmission Star, 50/125 Micrometer Fiber, Graded Index Multi-Mode Bidirectional (Metric).
DOD-C-24621/3	-	Couplers, Fiber Optic Cable, Cable Splitter, Passive, Glass-Pigtailed Output; Type II, 820 to 910 Nanometers Wavelength Range, 20 X 20 Port Transmission Star, 100/140 and 50/125 Micrometer Fiber.
DOD-C-24621/4	-	Couplers, Fiber Optic Cable, Cable Splitter, Passive, Glass-Pigtailed Output; Type II, 820 to 910 Nanometers Wavelength Range, 5 X 5 Port Transmission Star, 50/125 and 100/140 Micrometer Fiber.
MIL-C-24621/5	-	Coupler, Fiber Optic Cable, Cable Splitter, Passive, Glass-Pigtailed Output, Type II, 820 to 910 Nanometer Wavelength Range, 2 X 2 Symmetrical Transmission Star, 62.5/125 Micrometer Fiber, Bidirectional.
MIL-S-24623	-	Splice, Fiber Optic Cable, General Specification For (Metric).
MIL-C-28876	-	Connectors, Fiber Optic Environmental Resisting (For Navy Shipboard Application) General Specification For.
DOD-F-49291	-	Fiber, Optional (Metric) General Specification For.
MIL-C-55330	-	Connectors, Electrical and Fiber Optic, Packaging of.
MIL-C-83522	-	Connectors, Fiber Optic, Single Terminus, General Specification For.
DOD-C-85045	-	Cables, Fiber Optic, Ruggedized, Radiation Hardened (Metric).

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1344	-	Test Methods for Electrical Connectors.
DOD-STD-1678	-	Fiber Optic Test Methods and Instrumentation.
MIL-STD-2196	-	Glossary, Fiber Optics.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRY ASSOCIATION (EIA)

EIA-455-20	-	Measurement of Change in Optical Transmittance.
EIA-455-22	-	Ambient Light Susceptibility.
EIA-455-32	-	Fiber Optic Circuit Discontinuities.
EIA-455-36	-	Twist Test for Fiber Optic Connecting Devices.
EIA-455-49	-	Procedure for Measuring Gamma Irradiation Effects in Optical Fibers and Cables.
EIA-455-180	-	Measurement of the Optical Transfer Coefficients of a Passive Branching Device (Coupler).

(Application for copies should be addressed to the Electronic Industry Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Fiber optic coupler parts furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.3).

3.3 Material. The coupler, coupler connector caps, covers, storage receptacles, or other protective accessory hardware shall be constructed of material as specified herein or in the specification sheet. The material used in the coupler shall be as follows:

a. Resistant to corrosion as specified in requirement 15 of MIL-STD-454.

b. Inert to fungus (nonnutritive) as specified in requirement 4 of MIL-STD-454.

c. Nontoxic, nonflammable, and nonexplosive as specified in requirement 3 of MIL-STD-454, when exposed to heat, flame, or chemicals.

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d. Nonmagnetic under normal circumstances (with maximum micron (μ) less than or equal to 2.0 as determined by measurement with equipment conforming to MIL-I-17214).

e. Devoid of elemental mercury and asbestos.

f. Manufactured to good workmanship quality as specified in requirement 9 of MIL-STD-454.

3.3.1 Recovered materials. Unless otherwise specified herein, all materials incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification. Reground polymers are acceptable materials.

3.3.2 Steel parts. All exposed corrosion resistant steel parts (as specified in QQ-S-766) of the coupler assembly shall have a passivated finish which permits the attainment of a surface finish condition in accordance with high workmanship standards and shall be compatible with external coatings and platings of the type and color as specified in MIL-S-5002.

3.3.3 Dissimilar metal junctions. Where dissimilar metals are used in intimate contact with one another, suitable protection against electrolytic corrosion shall be provided as specified in MIL-STD-889.

3.3.4 Nonmetallic materials. Nonmetallic materials used in the construction of couplers shall not be degraded when exposed to the specified environmental conditions.

3.3.5 Solvents, adhesives, and cleaning agents. If any chemicals or cements are used in bonding of internal coupler parts or the fiber/terminus securing-polishing processes, incompatibility shall not exist between the materials employed such that degradation of these materials shall result from in-service use. Any epoxy or adhesives used in coupler construction shall conform to requirement 23 of MIL-STD-454. Additionally, the coupler manufacturer shall supply a list of recommended solvents, adhesives, and cleaning agents.

3.3.6 Sealing compounds. Sealing compounds shall not flow at the maximum specified storage temperature nor crack at the minimum specified storage temperature.

3.3.7 Fiber material. The optical fiber material of the coupler shall be in accordance with DOD-F-49291.

3.3.8 Fungus resistance. Materials and finishes shall be certified that they meet requirement 4 of MIL-STD-454. Couplers that are not in accordance with requirement 4 of MIL-STD-454 for fungus inert materials shall meet grade I classification of MIL-STD-810, method 508. If certification cannot be made, two additional sample units shall be tested in accordance with 4.8.20.

3.4 Design and construction. The coupler shall be mechanically and optically compatible with those sizes and types of fiber optic cables as specified in DOD-C-85045. Where cables or connectors are employed as part of the coupler assembly on the outside of the coupler enclosure, the cables or connectors shall meet the applicable fiber optic cable or connector requirements. The coupler shall include optical pigtails or connectors for interconnection within an optical link as input and output ports, an optical coupling junction, and a protective housing.

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3.4.1 Component design and construction. Couplers and associated accessories constructed to this specification shall withstand normal handling incident to installation and in-service maintenance.

3.4.2 Housing. The coupling junction shall be enclosed in a sealed watertight housing as specified herein. The housing for couplers shall include hardware for mounting.

3.4.3 Dimensions. The dimensions and dimensional tolerances for these couplers and associated accessories shall conform to the limits as specified (see 3.1).

3.4.4 Mass. The mass of the couplers and associated accessories shall conform to the limits when specified (see 3.1 and 4.6.3).

3.4.5 Seals. Grommets, O-rings, interfacial seals, boots, gaskets, or other sealing devices, as needed by the particular coupler design, shall accomplish their intended purpose and meet all test requirements as specified herein.

3.4.6 Optical junction sealing. Optical junctions shall be sealed against moisture, contamination, and mechanical damage as specified herein.

3.4.7 Protective covers. Coupler-connector receptacles or pig-tailed cable connectors shall be provided with a protective cover to prevent exposure and damage from ordinary handling, moisture, and dust when the mating connector is not attached. Protective covers shall be molded or machined from plastic or nonferrous material suitable for fitting the appropriate coupler part. Each plug or receptacle connector shall be fitted with a slide-on or screw-on protective cap on the front end of each assembly, as applicable, prior to unit packaging to protect termini and seal surfaces from entry of foreign matter, and to protect the surfaces from mechanical damage.

3.4.8 Maintainability. The coupler shall require no preventive maintenance.

3.5 Toxic and hazardous products and formulations. Materials and combinations of materials used in construction shall have no adverse effect on the health of personnel or equipment when used for its intended purpose. Where new or questionable materials are being considered for use, the contractor shall furnish to the cognizant government agency the toxicological data and formulations required to evaluate the safety of the materials for the proposed use.

3.6 Optical performance.

3.6.1 Signal discontinuity (see 6.4.8). When tested in accordance with 4.7.1, discontinuities shall not occur in the coupler's optical circuit signal. A discontinuity is considered a reduction in signal strength of more than 0.3 dB for a duration of 50 μs or more.

3.6.2 Ambient light susceptibility (see 6.4.1). When tested in accordance with 4.7.2, the optical power of the light from the nonenergized optical ports (after accounting for cable and optical junction losses between the coupler device and the detector) shall be less than -50 dBm.

3.6.3 Transmission matrix (see 6.4.9). Each coupler shall be tested for quality of performance according to the test configuration described by 4.7.3. Each value of P_{IJ} must be determined, and found to agree with the value specified (see 3.1) (within the allotted tolerance) for each specified wavelength range. The quantities P_{IJ} of this matrix when combined and substituted in the proper mathematical expression also define all the port-to-port and total insertion loss (see 6.4.4) values for a particular coupler.

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3.6.4 Change in optical transmittance. Each P_{IJ} of the transmittance matrix associated with one wavelength of operation of the coupler shall not change more than 0.3 dB when subjected to mechanical and environmental stress (see 4.7.4). For couplers with more than eight ports total, a minimum of seven ports shall be monitored. Nonadjacent ports shall be used as specified (see 3.1).

3.6.5 Wavelength sensitivity. When tested in accordance with 4.7.5, the change in optical transmittance shall be less than 0.3 dB over the wavelength spectrum specified (see 3.1).

3.6.6 Polarization sensitivity. When tested in accordance with 4.7.6, the normalized optical transmittance shall not deviate more than 5 percent over 180° of rotation for single mode couplers.

3.7 Environmental requirements.

3.7.1 Temperature life. When tested in accordance with 4.8.1, couplers shall not exhibit visual evidence of dimensional change, opening of seals, cracking or crazing of components or finishes, identification marking impairment, fusion or seizure of mating parts, leakage of waterproofing compounds or other effects detrimental to coupler operation. The change in optical transmittance shall not exceed the allowable limits after exposure. Also, the coupler must meet the requirements for ambient light susceptibility (see 3.6.2).

3.7.2 Fluid immersion. When tested in accordance with 4.8.2, a posttest visual inspection shall reveal no cracks, splits, or voids in the coupler assembly nor shall it become gummy or tacky. Visual inspection of the test coupler shall reveal no swelling or softening of material, no loss of sealing capability or identification marking, and no discoloration or other effects detrimental to the intended use of these optical components.

3.7.3 Temperature/humidity cycling. When tested in accordance with 4.8.3 for accelerated environment exposure of high humidity and cyclic temperatures, internal and external parts shall not swell or otherwise degrade such that coupler performance is impaired. A posttest visual inspection shall reveal no component softening, surface damage, or identification marking impairment. The change in optical transmittance shall not exceed the allowable limits as specified (see 3.1) during and after this exposure (see 3.6.4).

3.7.4 Water pressure. When tested in accordance with 4.8.4, visual inspection of the test coupler shall reveal no penetration of indicator dye into the sealed region of the couplers. The change in optical transmittance shall not exceed the allowable limits as specified (see 3.1) after this exposure (see 3.6.4).

3.7.5 Identification marking durability. When tested in accordance with 4.8.5, the coupler identification marking shall remain legible and intact.

3.7.6 Impact. When tested in accordance with 4.8.6, couplers and associated connectors shall not be visibly damaged or otherwise rendered unfit for operational use. Also, the change in optical transmittance shall not exceed the allowable limits as specified (see 3.1) before and after each event of this test (see 3.6.4).

3.7.7 Crush. When tested in accordance with 4.8.7, the coupler body shall show no evidence of broken parts or accessories, loss of optical continuity, or damage to seals. The change in optical transmittance shall not exceed the allowable limits as specified in 3.6.4 during and after exposure.

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3.7.8 Axial cable loading.

3.7.8.1 Axial cable loading/tension (type I couplers only). When tested in accordance with 4.8.8.1, the change in optical transmittance shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure (see 3.6.4). The posttest visual inspection shall reveal no cracking, splitting, regions of localized tearing, or deformation.

3.7.8.2 Axial cable loading/compression (type I couplers only). When tested in accordance with 4.8.8.2, axial movement of the pigtail cable relative to the enclosure shall be no more than half the cable diameter. A posttest inspection of the coupler shall reveal no physical damage to the seals. The change in optical transmittance shall not exceed the allowable limits as specified (see 3.1) at any time during and after this exposure (see 3.6.4).

3.7.9 Fiber dynamic strength. When tested in accordance with 4.8.14, the minimum fiber dynamic strength shall be 350 MPa. There shall be no fiber to coupler seal damage or distortion or bending of coupler parts. The coupler shall meet the requirements of and 3.6.4 during and after the test.

3.7.10 Enclosure bending moment. When tested in accordance with 4.8.9, couplers shall sustain no permanent deformation. The change in optical transmittance shall not exceed the allowable limits as specified at any time during and after this exposure (see 3.6.4).

3.7.11 Cable twist (type I couplers only). When tested in accordance with 4.8.10, a posttest visual inspection of the coupler and cable assembly shall reveal no loss of environmental sealing, splitting, or other impairment due to the torsional bending exposure. The change in optical transmittance shall not exceed the allowable limits as specified at any time during and after this exposure (see 3.6 4).

3.7.12 Cable seal flexing (type I couplers only). When tested in accordance with 4.8.11, coupler strain relief mechanisms shall prevent loss of environmental sealing or other damage which may impair the coupler operation.

3.7.13 Mechanical shock. When tested in accordance with 4.8.12, coupler assemblies shall not be damaged and there shall be no loosening of parts. There shall be no discontinuities during the test and the change in optical transmittance shall be within the limits specified (see 3.6.1 and 3.6.4).

3.7.14 Thermal shock. When tested in accordance with 4.8.13, a posttest visual inspection shall reveal no softening, surface damage, or identification marking impairment. The test couplers shall reveal no evidence of part dimensional change, leakage of waterproofing compounds, or other apparent loss of sealing capability, coupling-thread binding or other evidence of mating or unmating incapability, and no other damage detrimental to the operation of the coupler. The change in optical transmittance shall not exceed the allowable limits as specified before and after this exposure (see 3.6.4).

3.7.15 Operating temperature. When tested in accordance with 4.8.15, a posttest visual inspection of the coupler shall reveal no defects or damage. The coupler body and pigtail cable leads shall not show evidence of temperature susceptibility in regards to this test. The coupler shall survive and show no signs of mating or unmating incompatibility, leakage or waterproof sealing degradation, swelling, stretching or shrinking, or surface identification marking impairment. The change in optical transmittance shall not exceed the allowable limits as specified during this exposure (see 3.6.4).

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3.7.16 Vibration. When tested in accordance with 4.8.16, the coupler shall exhibit no loosening of parts, relative motion between parts, or other damage which can produce physical distortion or wear and may result in fatigue of the mechanical parts or failure of the coupler. The change in optical transmittance and discontinuity shall not exceed the allowable limits as specified before, during and after this exposure (see 3.6.1 and 3.6.4).

3.7.17 Salt spray (corrosion). When tested in accordance with 4.8.17, no visible evidence of corrosion should be seen on the external or associated coupler parts.

3.7.18 Flammability. When couplers are tested in accordance with 4.8.18, burning after removal of the applied flame shall extinguish within the time specified by test condition C, method 1012 of MIL-STD-1344. Any observed afterglow shall cease within the time specified by test condition C, method 1012 of MIL-STD-1344. Also, there shall be no dripping which will cause ignition of flammable material placed below the flame's contact point with coupler. There shall be no violent burning or explosive type fire during and after flame application. The change in optical transmittance shall not exceed the allowable limits as specified in 3.6.4 during and after exposure.

3.7.19 Nuclear radiation resistance. Nuclear vulnerability requirements for gamma radiation shall be as specified in a classified appendix to the individual specification sheet (see 4.8.19). The coupler shall meet the requirements of 3.6.4 during and after the test.

3.8 Color. The external color of the couplers shall be as specified (see 3.1 and 6.2).

3.9 Marking. Unless otherwise specified (see 6.2), the finished coupler shall be identified by a marking applied to the outer surface of the coupler or visible through the outer surface. The characteristic and method of identification marking shall conform to the requirements of MIL-STD-1285. The identification marking characters shall be 2 millimeters in height unless otherwise specified (see 3.1). The identification marking shall be permanent and legible in accordance with the marking requirements of MIL-STD-1285. In particular, the identification marking must also include the words "Fiber Optic Coupler". Identification marking shall be applied with the vertical axis of the printed characters perpendicular to the longest dimension of the coupler housing. All characters shall be expected to remain legible for the 20-year expectancy lifetime of the coupler. Except as otherwise specified, the coupler accessory parts shall also be identified by a legible and permanent marking applied in accordance with MIL-STD-1285.

3.10 Workmanship. Couplers and accessories shall be free of manufacturing flaws that would degrade performance, inhibit proper connection to interfacing elements, and otherwise yield an inferior product. All details of workmanship shall be in accordance with high grade fiber optic technology manufacturing practice. The following are minimal considerations and are not intended to exclude other pertinent workmanship requirements deemed necessary by the contractor.

a. Outer surface shall be free of rust, burnt areas, abrasions, holes, roughhead areas, bulges, thin spots, and indentations.

b. Inner parts shall be free of fractures, holes, discontinuities, bulges, and thin spots.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for inspection. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of DOD-STD-1678.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the qualifying activity, on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample. The number of sample units comprising a sample of couplers to be subjected to qualification inspection shall be eight. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production.

4.4.2 Inspection routine. Sample units shall be subjected to the qualification inspection specified in table I in the order shown.

4.4.3 Failures. Any failures shall be cause for refusal to grant qualification.

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TABLE I. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical inspection	3.3, 3.4, 3.4.1, 3.4.2, 3.4.3, 3.9	4.6.2
Ambient light susceptibility	3.6.2	4.7.2
Transmission matrix	3.6.3	4.7.3
Wavelength sensitivity	3.6.5	4.7.5
Polarization sensitivity	3.6.6	4.7.6
Group II (2 samples)		
Temperature life	3.7.1	4.8.1
Nuclear radiation resistance	3.7.19	4.8.19
Identification marking durability	3.7.5	4.8.5
Fluid immersion	3.7.2	4.8.2
Group III (2 samples)		
Temperature/humidity cycling	3.7.3	4.8.3
Crush	3.7.7	4.8.7
Identification marking durability	3.7.5	4.8.5
Impact	3.7.6	4.8.6
Water pressure	3.7.4	4.8.4
Group IV (2 samples)		
Axial cable loading/tension	3.7.8.1	4.8.8.1
Axial cable loading/compression	3.7.8.2	4.8.8.2
Fiber dynamic strength	3.7.9	4.8.14
Enclosure bending moment	3.7.10	4.8.9
Cable twist	3.7.11	4.8.10
Cable seal flexing	3.7.12	4.8.11
Mechanical shock	3.7.13	4.8.12
Thermal shock	3.7.14	4.8.13
Operating temperature	3.7.15	4.8.15
Vibration	3.7.16	4.8.16
Salt spray (corrosion)	3.7.17	4.8.17
Flammability	3.7.18	4.8.18
Group V (2 samples)		
Fungus resistance	3.3.8	4.8.20

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4.4.4 Retention of qualification. To retain qualification, the contractor shall forward a report at least every 12 months to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (groups A and B) indicating as a minimum the number of lots that have passed and the number that have failed and the group they have failed. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of the tests performed for periodic inspection (group C) shall be forwarded at least every 36 months, including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 36-month period. If the summary of the test results indicates nonconformance with specification requirements, the corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month reporting period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the reporting period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the products (a representative product of each series, type, and class, etc.) to testing in accordance with the qualification inspection requirements and the reason for no production.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all the couplers of the same type, optical characteristics, and protective enclosure or coating produced under essentially the same conditions and offered for inspection during a period of 1 month.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, and shall be made on the same set of products, in the order shown.

4.5.1.2.1 Sampling plan. Tests shall be performed on 100 percent of the product supplied under this specification. There shall be no failures.

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TABLE II. Group A inspection.

Inspection 1/	Requirement paragraph	Test method paragraph
Visual and mechanical inspection	3.3, 3.4, 3.4.1, 3.4.2, 3.4.3, 3.9	4.6.2
Mass	3.4.4	4.6.3
Transmission matrix	3.6.3	4.7.3
Polarization sensitivity	3.6.6	4.7.6

1/ The contractor may use in-process controls for these requirements.

4.5.1.2.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in table II as the final step of this production process, group A inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for group A. Test conditions shall be equal to or more stringent than those specified for group A tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. The manufacturer shall also make available to the government all records of all detail test data resulting from production tests.

f. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

g. Manufacturers shall show the correlation between data for the test requested and the test specified in this specification.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III in the order shown. They shall be performed on sample units that have been subjected to and have passed the group A inspection.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105. Unless otherwise specified herein, S-4 level inspection shall be used. There shall be no failures.

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TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Wavelength sensitivity	3.6.5	4.7.5
Thermal shock	3.7.14	4.8.13
Fiber dynamic strength	3.7.9	4.8.14
Axial cable loading		
Tension	3.7.8.1	4.8.8.1
Compression	3.7.8.2	4.8.8.2
Cable seal flexing	3.7.12	4.8.11
Operating temperature	3.7.15	4.8.15
Ambient light susceptibility	3.6.2	4.7.2

4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.3), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspections shall consist of the inspections specified in table IV in the order shown. Group C inspection shall be performed on sample units of each type and selected from inspection lots which have passed groups A and B inspections. Group C inspection samples shall be representative of production.

4.5.2.1.1 Sampling plan. Every 36 months, four sample units which have passed groups A and B inspections shall be selected.

4.5.2.1.2 Disposition of samples. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the contractor shall notify the qualifying activity of the failure and take corrective action on the materials and processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, and so forth, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections tests or the inspection test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-55330.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I (2 samples)		
Thermal shock	3.7.14	4.8.13
Enclosure bending moment	3.7.10	4.8.9
Impact	3.7.6	4.8.6
Mechanical shock	3.7.13	4.8.12
Vibration	3.7.16	4.8.16
Salt spray (corrosion)	3.7.17	4.8.17
Flammability	3.7.18	4.8.18
Group II (2 samples)		
Ambient light susceptibility	3.6.2	4.7.2
Cable twist	3.7.11	4.8.10
Temperature life	3.7.1	4.8.1
Nuclear radiation resistance	3.7.19	4.8.19
Fluid immersion	3.7.2	4.8.2
Temperature/humidity cycling	3.7.3	4.8.3
Crush	3.7.7	4.8.7
Identification marking durability	3.7.5	4.8.5
Water pressure	3.7.4	4.8.4

4.6 Inspection methods.

4.6.1 Equivalent test methods. The use of equivalent test methods is allowed subject to the following conditions:

a. The allowance of an equivalent method is specified in this specification.

b. The manufacturer has conducted both test methods during during qualification testing and has submitted complete test data to the preparing activity.

c. The preparinng activity has approved the use of that method by that manufacturer.

4.6.2 Visual and mechanical inspection. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed using 5X magnification to verify that the design, construction, physical characteristics, dimensions, marking, workmanship, and materials are in accordance with the applicable requirements.

4.6.3 Mass. The coupler's mass, including pigtails, shall be tested using scales of sufficient range and accuracy appropriate for the specified weight and tolerance. The coupler shall meet the requirements of 3.4.4.

4.7 Optical test methods. Unless otherwise specified, the optical properties test methods shall be performed at a wavelength of 1300 ±20 nm. Likewise, the fibers or pigtail fibers for interconnection are to be compatible in terms of diameter, frequency, and refractive index profile characteristics. For multimode fibers, a noncoherent light source shall be used. Light launch conditions shall be as specified in table V or equivalent. Initial launch conditions may be interrupted between tests.

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TABLE V. Light launch conditions.

Single mode	1 turn 30 mm mandrel
Multimode	70/70 or restricted or equivalent mandrel wrap

4.7.1 Signal discontinuity (see 3.6.1). Signal discontinuities shall be measured in accordance with EIA-455-32. A maximum of 8 ports shall be monitored during the test.

4.7.2 Ambient light susceptibility (see 3.6.2). The ambient light susceptibility shall be measured in accordance with EIA-455-22, with the exception that the output power in the "on"state shall be referenced to 1 milliwatt.

4.7.3 Transmission matrix (see 3.6.3). The requirements for determining the transfer coefficients of the transmission matrix shall be in accordance with EIA-455-180. Each matrix element P_{IJ} represents insertion loss from port I to port J. The power is measured at all output ports for each input port I in successive tests until all ports have been individually connected as input ports even for the case where I = J. Cladding mode stripping (see 6.4.3) of the input power is required. However, there shall be no cladding mode stripping of the output power signals. The transmission matrix elements include coupler connection losses. Tolerances may be associated with each of the specified matrix elements as specified (see 3.1). For multimode couplers, the transmission matrix shall be measured for both of the following launches. A uniform overfill launch and a 70/70 launch (see 6.4.9 and 6.4.3).

4.7.4 Change in optical transmittance (see 3.6.4). The change in optical transmittance shall be measured in accordance with EIA-455-20 or by an equivalent method.

4.7.5 Wavelength sensitivity. The operational optical wavelength range shall be tested in conjunction with the testing of the transmittance matrix. For each specified wavelength range, the transmittance matrix shall be tested using a source that spans the wavelength range. The source used shall be randomly polarized. Laser sources shall not be used. The wavelength range shall be scanned in 10 nanometer steps (see 3.6.5).

4.7.6 Polarization sensitivity (see 3.6.6). Polarization sensitivity shall be tested as follows: The optical input and output ports of the coupler under test (see figure 1) shall be connected to short optical cables in which the cladding mode stripping occurs before the input to the coupler and between the relevant output ports and the detectors. Unused coupler ports are appropriately capped to restrict ambient light acceptance. The nominal wavelength of the source shall be 1.300 µm for type 1 couplers and 1.550 µm for type 2 couplers. The spectral width of the source shall be at least 0.080 µm. The λ2 retardation plate shall be aligned just prior to the polarization sensitivity test such that an extinction ratio of at least 30 dB is measured. The polarizer shall have an extinction ratio greater than or equal to the extinction ratio of the λ2 retardation plate. The polarization sensitivity test shall be performed by rotating the λ2 retardation plate through 180°. Data shall be obtained for the change in normalized optical transmittance versus polarization. The normalized optical transmittance shall be calculated as follows:

$$P_{norm} = P_{test} / \Sigma P_{output}$$

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Where P_{test} shall be the output power of the coupler port with the maximum deviation during the test and ΣP_{output} shall be the sum of the powers of the output ports of the coupler. The polarization sensitivity shall be calculated as follows:

$$(P_{norm}) \text{ max } X \text{ 100 percent}$$

Where P_{norm} shall be the deviation from the average for the coupler port with the maximum deviation. The requirements of 3.6.3 and 3.6.6 shall be verified. An equivalent test method is allowed as specified in 4.6.1.

4.8 Environmental test methods.

4.8.1 Temperature life (see 3.7.1). Fully assembled coupler specimens shall be tested in accordance with method 108 of MIL-STD-202. The exposure time shall be 96 hours. The exposure temperature shall be +110°C. The change in optical transmittance shall be measured after the test (see 3.6.4).

4.8.2 Fluid immersion (see 3.7.2). When specified (see 3.1), the coupler body shall be tested in accordance with method 1016 of MIL-STD-1344. All assemblies shall be subjected to all of the fluids, temperatures, and immersion times as specified below. Each assembly shall be completely dried after each immersion. Following the test, assemblies shall be examined for compliance to 4.6.2. Additional fluids will be tested when specified. After each test exposure, the assemblies shall be inspected externally for visible damage.

Fluid	Temperature (°C)	Immersion time (hours)
Isopropyl alcohol (see TT-I-735)	20 - 25	24
Automobile gas (see MIL-G-3056)	20 - 25	24
Hydraulic fluid (see MIL-H-5606)	48 - 50	24
Turbine fuel (see MIL-T-5624)	20 - 25	24
Lubricating oil (see MIL-L-23699)	73 - 77	24
Lube oil (see MIL-L-17331)	73 - 77	24
Coolant	20 - 25	24
Sea water	20 - 25	24

4.8.3 Temperature/humidity cycling (see 3.7.3). Coupler assemblies shall be tested in accordance with method 4030 of DOD-STD-1678. Cleaning of the unmated coupler connector junctions prior to optical testing is permissible, if necessary. The change in optical transmittance shall be measured during and after the test.

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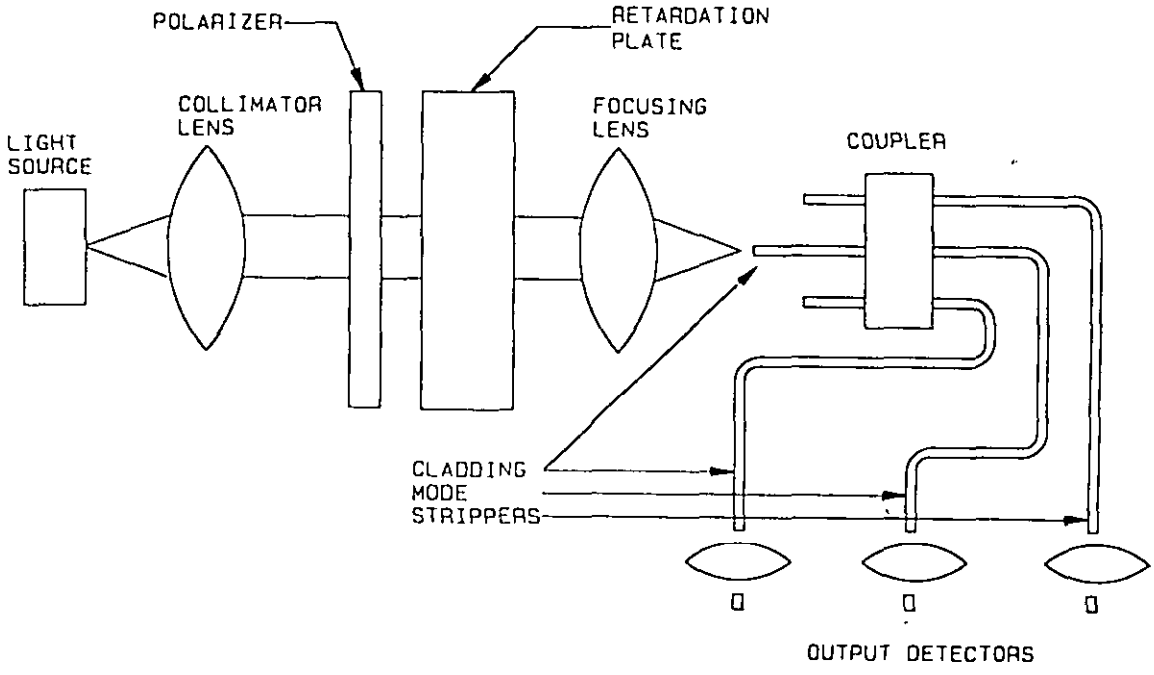


FIGURE 1. Polarization sensitivity test.

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4.8.4 Water pressure (see 3.7.4). Mated coupler cable assemblies shall be tested for water pressure susceptibility as follows:

a. The assemblies shall be immersed in an aqueous dye penetrant solution to a depth of 10.4 meters for a period of 48 hours.

b. The solution temperature shall be maintained between +10°C and +35°C during the exposure period.

c. The dye concentration shall visibly indicate liquid penetration.

d. After test exposure, the coupler assemblies shall be externally cleaned, externally inspected, mated, optically tested, and then internally inspected for dye penetration into the coupler.

4.8.5 Identification marking durability (see 3.7.5). The marking durability test shall be accomplished as described herein. The marking durability tester shall be designed to firmly hold the coupler with the marked face fully exposed. The tester shall be capable of rubbing a small cylindrical steel mandrel, 1.0 ±0.1 millimeter in diameter, repeatedly over the exposed surface of the coupler, as held in such a position that the longitudinal axes of the mandrel and the marking are at right angles to each other with their surfaces in contact. One kilogram mass affixed to a jig above the mandrel shall control the force exerted normal to the marking on exposed surface of the coupler. A motor driven mechanism and counter shall be used to deliver an accurate number of abrading strokes in a direction parallel to the direction of the marking. The length of the stroke shall be 10 millimeters and the frequency shall be 120 strokes (60 stroking cycles) per minute. The coupler shall be subjected to 600 strokes (300 stroking cycles).

4.8.6 Impact (see 3.7.6). Unmatted coupler body shall be tested in accordance with the following method. A total of three drops on each assembly shall be made from a height of 1 meter above a flat solid steel plate of thickness not less than 12.70 mm (.50). Each fall shall be oriented such that a coupler surface impacts the floor approximately normal to a different principal axis of the coupler geometry. Three different faces suffer impacts in the three successive and different drop tests where the coupler body is rotated by an angle of 90° relative to its reference (initial) position between each free fall. The coupler shall not be dropped on a face containing a connector, pigtail, cable, or any other device. The change in optical transmittance shall be monitored after the test (see 3.6.4).

4.8.7 Crush (see 3.7.7). When specified, couplers shall be tested in accordance with method 2008 of MIL-STD-1344 except that the test sample shall be a coupler. The test load shall be 1250 newtons. The change in optical transmittance shall be monitored during and after the test.

4.8.8 Axial cable loading.

4.8.8.1 Axial cable loading/tension (see 3.7.8.1). The axial cable loading test shall be in accordance with method 2009 of MIL-STD-1344 and the procedure described herein:

a. The coupler shall be securely mounted in the normal manner.

b. All cables on the most populated face shall be mechanically loaded in tension to one-half of the maximum tensile load for the pigtail cable or 100 newtons, whichever is less.

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c. Apply the tensile load at the rate of about 2000 newtons per minute, and hold the maximum load for 1 minute.

d. Remove the tensile load.

e. Visually inspect the coupler using 3X magnification.

f. The change in optical transmittance shall be measured during and after the test (see 3.6.4).

4.8.8.2 Axial cable loading/compression (see 3.7.8.2). The axial compressive load test shall be performed as described herein. If all parts of the coupler are essentially the same, only the input and output ports used for the insertion loss measurement need to be tested.

a. The test cables forming a part of the assembled coupler shall be gripped over a length equal to at least three cable diameters, starting one cable diameter from the coupler enclosure or the strain relief mechanism if it is external to the enclosure. The coupler enclosure shall be gripped so as not to alter any of its mechanical properties.

b. The gripped cable and coupler enclosure shall be forced together along the direction of the cable axis where the cable enters the coupler assembly. The force magnitude is a function of pigtail cable diameter as listed in table VI. Loading and unloading are such that the coupler is compressively loaded and unloaded linearly in time, with the loading and unloading times each less than 10 seconds. The time spent under full compressive load shall be at least 1 minute. Tensile loads shall not be applied to the pigtail cables.

TABLE VI. Force magnitude versus pigtail cable diameter.

Nominal cable diameter (mm)	Compressive axial force (newton)
0 to 3	10
3 to 6	20
6 to 10	50
10 to 20	100
20 and larger	200

c. Any test fixture can be used provided it adequately grips the coupler enclosure and cable, allows control and measurement of the applied compressive force, and allows visual observation of the region where the cable meets the coupler assembly. The test fixture must also allow for the performance of all optical tests called for while the assembly is subject to the compressive axial load.

d. With load applied, measure the distance between the fixture clamp and the enclosure or strain relief mechanism and record the insertion loss.

e. Remove the load.

f. Visual observation and inspection shall be made with 3X magnification.

g. The change in optical transmittance shall be measured during and after the test (see 3.6.4).

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4.8.9 Enclosure bending moment (see 3.7.10). Mated coupler connector assemblies shall be tested in accordance with the following procedure which checks proper operation of coupler under deformation due to applied torque, a condition which can occur as a result of uneven stresses from mounting or installation. This test considers as a reference, the practical case of a coupler body 50 mm (1.97 inches) in length, mounted on a flat surface made uneven by a 0.5 mm (.02 inch) shim. This small shim, which is used to model a typical convex surface of attachment in shipyard construction, is to be situated midway underneath the longest extent of the coupler body. The coupler shall be balanced in a stable position atop a small metal block (or shim). The width of the fulcrum (a bar of shim), upon which the coupler rests shall be greater than or equal to 0.2 mm (.008 inch) (but never with the contact span less than 2 mm (.08 inch) under any circumstances). The fulcrum shall be secured against a smooth, level foundation offering stiff support. Install the coupler according to the manufacturer's recommended mounting instructions, but with the shim centered underneath the coupler. In the event that such instructions are not available from the manufacturer, the sample description below pertains to a coupler bearing four screw holes for attachment to a flat base, as illustrated in figure 2, configuration A. The four screws are to be tightened gradually in quarter turn clockwise increments once the coupler body becomes loosely clamped against the fulcrum during this fastening process. The four screws shall be attended by small quarter turn twists one after another in cyclic permutations such that all four bolts become progressively tighter. The force applied by the mounting screws of the coupler shall be increased until the loading effect is sufficient to deflect the far edges downward to the level of the foundation (but only to the degree that the tips of the coupler body just barely touch the foundation below). This can be determined electronically by sensing the moment that an electrical conduction path is established while the coupler gradually becomes more tightly fastened. The insertion loss of the "bent" coupler is measured at this time and compared with the maximum allowed value, as specified (see 3.1). Each of the four screws (fasteners) shall be tightened. The test configuration and definition of bending moment angle θ_{BM} in terms of the deformed coupler body are shown on figure 2, configuration B. Note: The combination of the shim and the coupler, as shown, defines a reference angle (of deformation) of 1.15°. The change in optical transmittance of the deformed coupler under stress/strain is then measured to verify that it (including any induced attenuation caused by this test) still conforms to values less than the maximum allowed change in optical transmittance as specified (see 3.1 and 3.6.4).

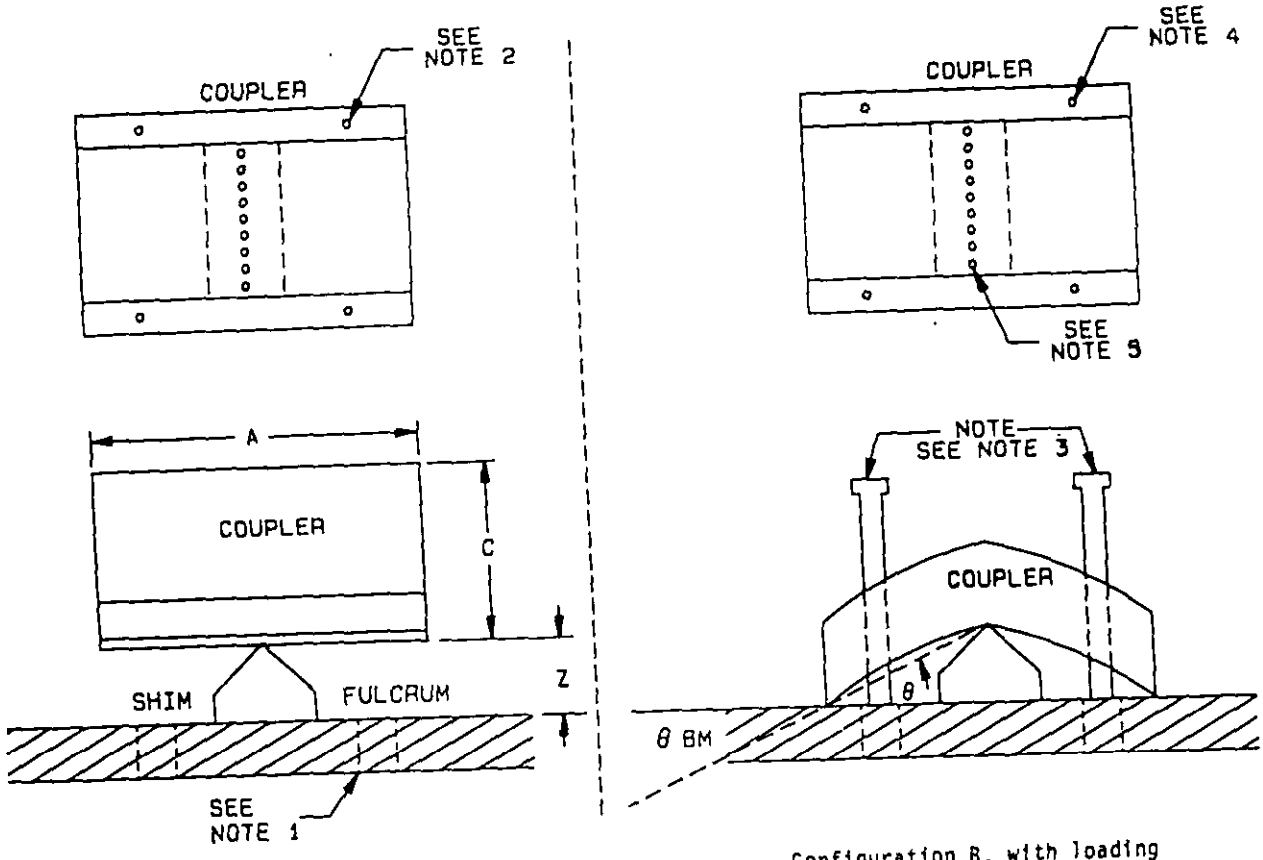
NOTE: The fulcrum bar (or shim) shall be of ample length to extend across the full width of the coupler body. Both the pivot piece of shim (fulcrum bar) and floor support beneath the fulcrum shall be of rigid, inelastic and smoothly finished material. The proper height Z (of shim) required for a fulcrum to induce a 1.15° deflection angle on a coupler of length A is (according to the geometry of figure 2, configuration B):

$$Z = .50\ A \tan (1.15^\circ)$$
$$Z = 0.01\ A$$

4.8.10 Cable twist (for type I couplers)(see 3.7.11). Coupler assemblies shall be tested in accordance with EIA-455-36 for a total of 50 twisting cycles. The minimum tensile load shall be 45 newtons. The change in optical transmittance shall be measured during and after the test (see 3.6.4). The coupler shall be inspected under 3X magnification.

4.8.11 Cable seal flexing (for type I couplers)(see 3.7.12). Coupler assemblies shall be tested in accordance with method 2017 of MIL-STD-1344.

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NOTES:
1. Predrilled holes to implement flange type mounting of coupler over area with shim.
2. Four preexisting holes in flange edge provided by manufacturer for easy, rigid mounting of coupler body.
3. Screw-driven torque shall be applied at all four points of mounting via flange.
4. Loading points of coupler after tight fastening.
5. Shim strip or fulcrum "lies" underneath middle zone of coupler.

FIGURE 2. Enclosure bending moment test configurations.

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4.8.12 Mechanical shock (see 3.7.13). Mated coupler cable assemblies shall be tested in accordance with the methods of MIL-S-901, grade A, class I, type B. When specified, the coupler shall be tested to MIL-STD-883, method 2002, test condition B instead of MIL-S-901. Discontinuity shall be monitored during the test (see 3.6.1).

4.8.13 Thermal shock (see 3.7.14). Thermal shock testing shall be accomplished in accordance with method 1003 of MIL-STD-1344. Cleaning of optical contacts of coupler connectors prior to the test exposure shall be permitted.

a. Test method cycles shall be as indicated in test condition A, except that the low and high exposure temperatures shall be the low and high nonoperating temperatures.

b. The time duration of each half cycle temperature exposure (steps 1 and 3) shall be the minimum (30 minutes) as specified in method 1003 of MIL-STD-1344. The maximum time condition (5 minutes) is specified for the interim temperature of steps 2 and 4.

c. Pretest and posttest measurements of the couplers dimensions shall be made and reported.

d. Visual inspection of the coupler shall be accomplished using 3X magnification.

e. The change in optical transmittance shall be measured after the test.

4.8.14 Fiber dynamic strength (see 3.7.9). The fiber pigtail dynamic strength shall be tested as follows: The fiber pigtail shall have an axial tensile load applied up to the load specified at an angle of 45° to the normal (see figure 3). With the load applied, the pigtail shall be rotated through one rotation (360°). The change in optical transmittance shall be monitored during and after the test (see 3.6.4). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.2. The coupler shall meet the requirements specified in 3.7.9).

4.8.15 Operating temperature (see 3.7.15). The operating temperature test shall be measured in accordance with method 4010 of DOD-STD-1678 over the operating temperature extremes. The change in optical transmittance shall be measured during and after the test (see 3.6.4).

4.8.16 Vibration (see 3.7.16). All couplers with mated cable assemblies shall be tested in accordance with test conditions D and B (at a level of 10 g rms) of method 204 of MIL-STD-202. Optical discontinuity shall be measured during the test (see 3.6.1).

4.8.17 Salt spray (corrosion) (see 3.7.17). Mated coupler cable assemblies shall be tested in accordance with test condition A of method 1001 of MIL-STD-1344. After test exposure, the assemblies shall be externally cleaned and inspected for evidence of corrosion.

4.8.18 Flammability (see 3.7.18). Flammability of couplers and associated accessories shall be tested under test condition C in accordance with method 1012 of MIL-STD-1344. The change in optical transmittance shall be measured during and after the test (see 3.6.4).

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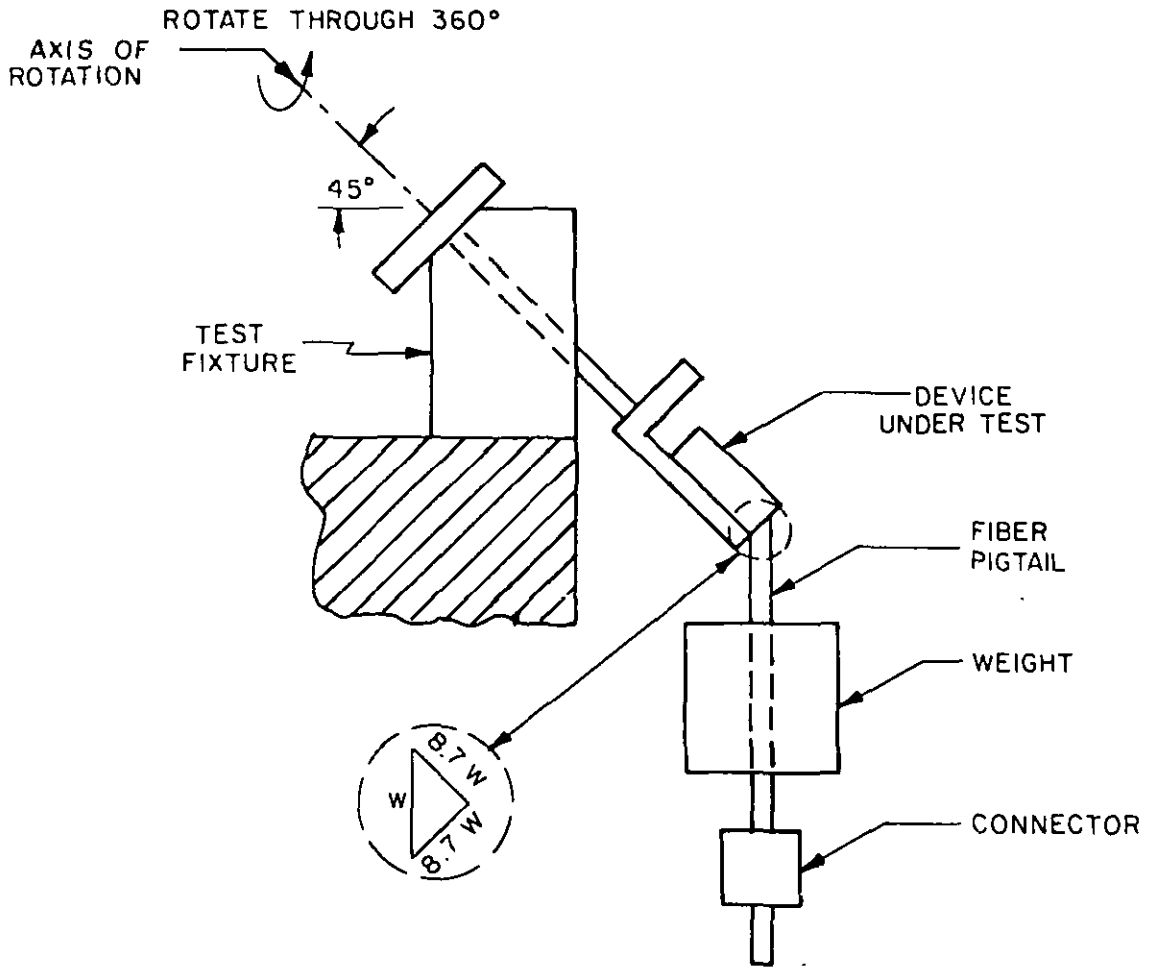


FIGURE 3. Fiber dynamic strength test setup.

4.8.19 *Nuclear radiation resistance (see 3.7.19).* Coupler assemblies shall be tested in accordance with EIA-455-49. The tests shall be performed at a wavelength of 1300 ±20 nm at the low operating temperature, at room temperature, and at the high operating temperature. The dose rate shall be 5 rad/s. The change in optical transmittance shall be measured during and after the test (see 3.6.4 and 3.7.19).

4.8.20 *Fungus resistance.* Couplers that do not meet the requirements of fungus materials in accordance with requirement 4 of MIL-STD-454 shall be tested for exposure to fungus in accordance with method 508 of MIL-STD-810 (see 3.3.8).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fiber optic couplers covered by this specification are intended for use in any application where their performance characteristics are required. The couplers are suitable for installation on all military systems within the limitation of their specified performance requirements.

6.1.1 Primary design considerations. Primary design shall be as follows:

a. Reliable for operation under conditions of temperature, salt spray, moisture, and vibration as described herein and in accordance with requirement 35 of MIL-STD-454.

b. Lightweight and compact.

c. Maximum compatibility shall exist between couplers, connector receptacles, and interconnecting cables without modification thereto or use of special adapters.

d. Maximum ease of installation.

e. Satisfactory identification for ease of requisitioning.

f. The coupler shall provide reliable operation under the environmental and mechanical requirements specified herein for the intended lifetime of the coupler which shall not be less than 20 years.

g. The coupler shall require no preventive maintenance.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Specification sheet number, title, and date.

c. PIN.

d. External color required (see 3.8).

e. Exceptions, if any, to the optional provisions of this specification are as follows:

(1) Exceptions to product identification requirements (see 3.7.5), if applicable.

(2) Responsibility for inspection, if other than specified (see 4.1).

(3) Special preparation for delivery requirements, if applicable (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-24621 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 55Z3, DoD Standardization Program and Documents Division, Washington, DC 20362-5101; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center, DESC-EQP, Dayton, OH 45444-5287.

6.3.1 Provisions governing qualification SD-6. Copies of "Provisions Governing Qualifications" SD-6 may be obtained upon application to Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.4 Definitions. Definitions of terms shall be in accordance with MIL-STD-2196 and as stated below.

6.4.1 Ambient optical pickup. Ambient optical pickup refers to the coupling of ambient optical power incident upon the exterior of a coupling device into the optical signal channels of the coupling device. The purpose of an ambient optical link test is to verify the immunity of the optical coupling device to ambient radiation (optical interference).

6.4.2 Backscatter. Backscatter is the component of electromagnetic waves that are directed back towards their source when resolved along a line from the source to the point of scatter. In this specification, backscatter is measured as the ratio of the power in dB of the light deflected along a path opposite to the launch path.

6.4.3 Cladding mode stripping. Cladding mode stripping refers to a method of eliminating the high loss modes from a fiber optic input signal, before introduction into the optical component of interest, e.g., detector, coupler or waveguide. This can help to ensure detector measurements that represent practical systems operate at equilibrium levels.

6.4.4 Insertion loss (port-to-port). The insertion loss of an optical coupling device is the optical signal power loss in an optical channel caused by insertion of the optical coupler into the optical circuit containing that optical channel. Insertion loss defines a value for the attenuation of a particular optical path from input port J to output port I through the coupler. The test provides a means of investigating and quantifying possible performance degradation during or after environmental or mechanical test exposures by comparison against pretest values.

6.4.5 Isolation. Isolation is, in a fiber optic coupler, the extent to which optical power from one signal path is prevented from reaching another signal path.

6.4.6 Nonuniformly distributive couplers. A nonuniformly distributive coupler is a coupler in which an optical input to each input port of the coupler is not equally distributed among the output ports, and an optical input to each output port of the coupler is not equally distributed among the input ports.

6.4.7 Passive couplers. A passive coupler is an optical device which splits or divides an input signal at one port into two or more output signals (with each output signal extracted from different ports) without any modifications, modulation, or amplification of the input within the coupler. A passive coupler does not contain an optical source, detector, or photosensitive element of any type.

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6.4.8 Signal discontinuity. A signal discontinuity refers to a sharp, short duration interruption or drop-out in the optical signals transmitted by an optical coupling device. The purpose of a continuity test is to verify that the coupling device does not introduce any signal discontinuities when it is subjected to mechanical or environmental stress.

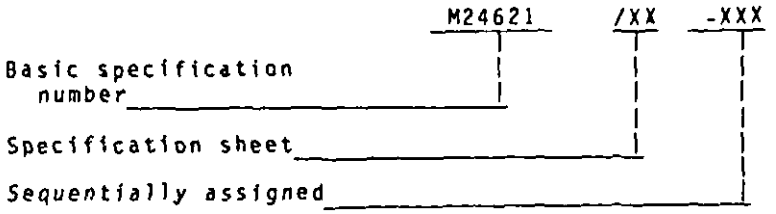
6.4.9 Transmission matrix. The transmission matrix characterizes the distribution of input energy from an input port I among all ports J. For an N X M port coupler, the matrix is as follows:

$$\begin{matrix} P_{11} & P_{12} & P_{13} & \cdot & \cdot & \cdot & P_{1L} \\ P_{21} & P_{22} & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ P_{L1} & \cdot & \cdot & \cdot & \cdot & \cdot & P_{LL} \end{matrix}$$

Where L = M + N. Each matrix element P_{IJ} (in dB) is the optical power level out of port J relative to the optical power level input to port I: $P_{IJ} = 10\log(P_J/P_I)$. The purpose of a transmission matrix test is to confirm the reputed transmission characteristics of the coupler (see EIA-455-180 for a more detailed definition). Transmission matrix elements include insertion loss, reflectance, backscatter, and isolation.

6.4.10 Uniformly distributive couplers. A uniformly distributive coupler is a coupler in which an optical input to each input port of the coupler is equally distributed among the output ports, and an optical input to each output port of the coupler is equally distributed among the input ports.

6.5 PIN. PIN's for couplers should contain only the following:



Example: M24621/01-001

6.6 Subject term (key word) listing:

Closure
Passive
Waveguide
Return loss
Splitter
Wavelength division multiplexing
Wavelength independent device
Optical transmittance

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Navy - AS, EC
Air Force - 11, 17, 19, 80, 99

User activities:
Navy - CG, MC

Preparing activity:
Navy - SH

Agent:
DLA - ES

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-24621B	6June 1991

3. DOCUMENT TITLE

COUPLERS, PASSIVE, FIBER OPTIC, GENERAL SPECIFICATION FOR (METRIC)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Technical point of contact(TPC) Walter Isengard, Code 51J1, (703) 602-0789	(1) Commercial (2) AUTOVON (AV) 332-0789
c. ADDRESS (Include Zip Code) Commander Naval Sea Systems Command AttN: 55Z33 Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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